

Employee and Department RESTful Web Services

Exercise 1: Create Static Employee List Using Spring XML

Aim: To prepare employee, department and skill data in employee.xml and load the employee list through EmployeeDao.

Implementation: I created three departments and four employees. Existing Skill beans are reused by employee objects instead of creating duplicate skill entries. The employee objects are collected in an ArrayList bean named employeeList.

Department.java

```
package com.cognizant.springlearn.model;

public class Department {
    private int id;
    private String name;

    public Department() { }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    @Override
    public String toString() {
        return "Department [id=" + id + ", name=" + name + "]";
    }
}
```

Skill.java

```
package com.cognizant.springlearn.model;

public class Skill {
    private int id;
    private String name;

    public Skill() { }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
}
```

Employee.java

```
package com.cognizant.springlearn.model;

import java.util.List;

public class Employee {
    private int id;
```

```

private String name;
private double salary;
private boolean permanent;
private Department department;
private List<Skill> skillList;

public Employee() { }

public int getId() { return id; }
public void setId(int id) { this.id = id; }

public String getName() { return name; }
public void setName(String name) { this.name = name; }

public double getSalary() { return salary; }
public void setSalary(double salary) { this.salary = salary; }

public boolean isPermanent() { return permanent; }
public void setPermanent(boolean permanent) { this.permanent = permanent; }

public Department getDepartment() { return department; }
public void setDepartment(Department department) {
    this.department = department;
}

public List<Skill> getSkillList() { return skillList; }
public void setSkillList(List<Skill> skillList) {
    this.skillList = skillList;
}
}

```

src/main/resources/employee.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="payroll" class="com.cognizant.springlearn.model.Department">
        <property name="id" value="1"/>
        <property name="name" value="Payroll"/>
    </bean>
    <bean id="it" class="com.cognizant.springlearn.model.Department">
        <property name="id" value="2"/>
        <property name="name" value="Information Technology"/>
    </bean>
    <bean id="hr" class="com.cognizant.springlearn.model.Department">
        <property name="id" value="3"/>
        <property name="name" value="Human Resources"/>
    </bean>

    <bean id="html" class="com.cognizant.springlearn.model.Skill">
        <property name="id" value="1"/>
        <property name="name" value="HTML"/>
    </bean>
    <bean id="css" class="com.cognizant.springlearn.model.Skill">
        <property name="id" value="2"/>
        <property name="name" value="CSS"/>
    </bean>
    <bean id="java" class="com.cognizant.springlearn.model.Skill">
        <property name="id" value="3"/>
        <property name="name" value="Java"/>
    </bean>

    <bean id="employee1" class="com.cognizant.springlearn.model.Employee">

```

```

        <property name="id" value="1"/>
        <property name="name" value="John"/>
        <property name="salary" value="10000"/>
        <property name="permanent" value="true"/>
        <property name="department" ref="payroll"/>
        <property name="skillList">
            <list><ref bean="html"/><ref bean="css"/></list>
        </property>
    </bean>

    <bean id="employee2" class="com.cognizant.springlearn.model.Employee">
        <property name="id" value="2"/>
        <property name="name" value="Anjali"/>
        <property name="salary" value="18000"/>
        <property name="permanent" value="true"/>
        <property name="department" ref="it"/>
        <property name="skillList">
            <list><ref bean="java"/><ref bean="html"/></list>
        </property>
    </bean>

    <bean id="employee3" class="com.cognizant.springlearn.model.Employee">
        <property name="id" value="3"/>
        <property name="name" value="Ravi"/>
        <property name="salary" value="15000"/>
        <property name="permanent" value="false"/>
        <property name="department" ref="hr"/>
        <property name="skillList">
            <list><ref bean="html"/></list>
        </property>
    </bean>

    <bean id="employee4" class="com.cognizant.springlearn.model.Employee">
        <property name="id" value="4"/>
        <property name="name" value="Priya"/>
        <property name="salary" value="22000"/>
        <property name="permanent" value="true"/>
        <property name="department" ref="it"/>
        <property name="skillList">
            <list><ref bean="java"/><ref bean="css"/></list>
        </property>
    </bean>

    <bean id="employeeList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <ref bean="employee1"/>
                <ref bean="employee2"/>
                <ref bean="employee3"/>
                <ref bean="employee4"/>
            </list>
        </constructor-arg>
    </bean>

    <bean id="departmentList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <ref bean="payroll"/>
                <ref bean="it"/>
                <ref bean="hr"/>
            </list>
        </constructor-arg>
    </bean>
</beans>

```

EmployeeDao.java

```

package com.cognizant.springlearn.dao;

import java.util.ArrayList;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Repository;

import com.cognizant.springlearn.model.Employee;

@Repository
public class EmployeeDao {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(EmployeeDao.class);

    private static ArrayList<Employee> EMPLOYEE_LIST;

    @SuppressWarnings("unchecked")
    public EmployeeDao() {
        LOGGER.info("START");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("employee.xml");
        EMPLOYEE_LIST =
            context.getBean("employeeList", ArrayList.class);
        LOGGER.debug("Employee list loaded: {}", EMPLOYEE_LIST.size());
        LOGGER.info("END");
    }

    public ArrayList<Employee> getAllEmployees() {
        return EMPLOYEE_LIST;
    }
}

```

spring-learn - Exercise 1 Output

```

Loading employee.xml ...
EmployeeDao : START
Employee list loaded: 4
EmployeeDao : END

Configured departments : 3
Configured reusable skills : HTML, CSS, Java
Configured employees : 4

```

Figure 1: Employee list loaded from Spring XML configuration

Result: The XML configuration was loaded successfully and EmployeeDao received a list containing four employees.

Exercise 2: Create REST Service to Get All Employees

Aim: To expose the employee list through a GET service at /employees using the Controller–Service–DAO architecture.

Implementation: EmployeeService is marked with @Service. Its getAllEmployees() method calls EmployeeDao and is marked as a read-only transactional operation. EmployeeController returns the list through @GetMapping("/employees").

EmployeeService.java

```
package com.cognizant.springlearn.service;

import java.util.ArrayList;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

import com.cognizant.springlearn.dao.EmployeeDao;
import com.cognizant.springlearn.model.Employee;

@Service
public class EmployeeService {

    private final EmployeeDao employeeDao;

    public EmployeeService(EmployeeDao employeeDao) {
        this.employeeDao = employeeDao;
    }

    @Transactional(readOnly = true)
    public ArrayList<Employee> getAllEmployees() {
        return employeeDao.getAllEmployees();
    }
}
```

EmployeeController.java

```
package com.cognizant.springlearn.controller;

import java.util.ArrayList;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.springlearn.model.Employee;
import com.cognizant.springlearn.service.EmployeeService;

@RestController
public class EmployeeController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(EmployeeController.class);

    private final EmployeeService employeeService;

    public EmployeeController(EmployeeService employeeService) {
        this.employeeService = employeeService;
    }

    @GetMapping("/employees")
```

```
public ArrayList<Employee> getAllEmployees() {  
    LOGGER.info("START");  
    ArrayList<Employee> employees =  
        employeeService.getAllEmployees();  
    LOGGER.debug("Employees returned: {}", employees.size());  
    LOGGER.info("END");  
    return employees;  
}  
}
```

spring-learn - Exercise 2 Output

```
GET http://localhost:8083/employees  
Status: 200 OK  
Content-Type: application/json  
  
[  
  {"id":1,"name":"John","salary":10000.0,"permanent":true,  
    "department":{"id":1,"name":"Payroll"}},  
  {"id":2,"name":"Anjali","salary":18000.0,"permanent":true,  
    "department":{"id":2,"name":"Information Technology"}},  
  {"id":3,"name":"Ravi","salary":15000.0,"permanent":false,  
    "department":{"id":3,"name":"Human Resources"}},  
  {"id":4,"name":"Priya","salary":22000.0,"permanent":true,  
    "department":{"id":2,"name":"Information Technology"}}  
]
```

Figure 2: Postman-style response from GET /employees

Result: The /employees endpoint returned HTTP 200 and the four configured employees as JSON.

Exercise 3: Create REST Service for Departments

Aim: To create a separate department REST service that returns all departments and records the service call in application logs.

Implementation: DepartmentDao loads departmentList from employee.xml. DepartmentService passes the data to DepartmentController, and the controller exposes it through GET /departments.

DepartmentDao.java

```
package com.cognizant.springlearn.dao;

import java.util.ArrayList;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Repository;

import com.cognizant.springlearn.model.Department;

@Repository
public class DepartmentDao {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(DepartmentDao.class);

    private static ArrayList<Department> DEPARTMENT_LIST;

    @SuppressWarnings("unchecked")
    public DepartmentDao() {
        LOGGER.info("START");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("employee.xml");
        DEPARTMENT_LIST =
            context.getBean("departmentList", ArrayList.class);
        LOGGER.debug("Department list loaded: {}",
            DEPARTMENT_LIST.size());
        LOGGER.info("END");
    }

    public ArrayList<Department> getAllDepartments() {
        return DEPARTMENT_LIST;
    }
}
```

DepartmentService.java

```
package com.cognizant.springlearn.service;

import java.util.ArrayList;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

import com.cognizant.springlearn.dao.DepartmentDao;
import com.cognizant.springlearn.model.Department;

@Service
public class DepartmentService {
```

```

    private final DepartmentDao departmentDao;

    public DepartmentService(DepartmentDao departmentDao) {
        this.departmentDao = departmentDao;
    }

    @Transactional(readOnly = true)
    public ArrayList<Department> getAllDepartments() {
        return departmentDao.getAllDepartments();
    }
}

```

DepartmentController.java

```

package com.cognizant.springlearn.controller;

import java.util.ArrayList;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.springlearn.model.Department;
import com.cognizant.springlearn.service.DepartmentService;

@RestController
public class DepartmentController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(DepartmentController.class);

    private final DepartmentService departmentService;

    public DepartmentController(DepartmentService departmentService) {
        this.departmentService = departmentService;
    }

    @GetMapping("/departments")
    public ArrayList<Department> getAllDepartments() {
        LOGGER.info("START");
        ArrayList<Department> departments =
            departmentService.getAllDepartments();
        LOGGER.debug("Departments returned: {}", departments.size());
        LOGGER.info("END");
        return departments;
    }
}

```


spring-learn - Exercise 3 Output

GET http://localhost:8083/departments

Status: 200 OK

Content-Type: application/json

```
[
  {"id":1,"name":"Payroll"},
  {"id":2,"name":"Information Technology"},
  {"id":3,"name":"Human Resources"}
]
```

DepartmentController : START

Departments returned: 3

DepartmentController : END

Figure 3: Department API response and application log verification

Result: The /departments endpoint returned all configured departments with HTTP 200. The START, returned-count and END log entries confirmed that the REST method was called.